



# Taxonomic, functional and phylogenetic diversity of ant communities in response to elevational gradients in the Qinling Mountains

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## Abstract

In the context of global warming, studying insect diversity in mountainous regions is crucial for predicting species distribution and guiding biodiversity conservation. To examine the patterns of ant diversity along elevational gradients, we conducted field surveys in the Qinling Mountains, a natural boundary between the warm temperate and subtropical zones in China. Ants were sampled from elevations ranging from 967 to 2029 m above sea level. We measured six functional traits related to diet, habitat preference, and foraging strategy of ants. Our study evaluated the influence of elevation on taxonomic, functional, and phylogenetic diversity, as well as ant community assembly. Our findings revealed differences in species composition of ant communities along elevations, and ant diversity tended to decrease with increasing elevation. The functional traits of ant communities at low and high elevation were similar (larger body sizes, wider heads, longer scapes, bigger eyes, longer mandibles and longer hind legs), while ants at the middle elevation showed opposite traits. Stochastic processes influenced ant communities at low elevation, and environmental filters played a primary role in shaping communities at middle and high elevation. The results suggest challenges for specialist ant species in dispersing across elevations, and provide valuable insights into ant ecological adaptation and diversity conservation in the Qinling Mountains.

**Keywords** Formicidae · Functional trait · Community assembly · Conservation

## Introduction

Climate warming, as one of the major causes of insect extinction, is impacting insect distribution (Menéndez et al. 2006; Sánchez-Bayo and Wyckhuys 2019). Some species tend to migrate towards cooler regions, like high-latitude regions or high-elevation habitats (Mené-

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ndez et al. 2006; Deutsch et al. 2008; Chen et al. 2009). Consequently, studying insect diversity and their community assembly mechanisms in mountains is a critical goal of community ecology, and provides valuable guidance for biodiversity conservation worldwide (Machac et al. 2011; Liu et al. 2018; de Castro et al. 2020). It is a general phenomenon that insect community species composition changes along elevational gradients (Hodkinson 2005). Insect species richness typically decreases or exhibits mid-elevation peaks with increased elevations (Fontanilla et al. 2019; Joseph et al. 2019; Silvestre et al. 2021; Henriques et al. 2022). Previous studies primarily focus on taxonomic diversity, failing to explain the ecological process of community structure response to elevations (Wei et al. 2001; He et al. 2003; Joseph et al. 2019; Xie et al. 2022). To comprehensively evaluate the variation of insect communities along elevational gradients, it is essential to simultaneously consider multiple aspects of taxonomic, functional, and phylogenetic diversity (Samways et al. 2020; Silvestre et al. 2021).

The functional traits, such as morphological, behavioral, physiological, and life-history characteristics, are the results of species interactions with biotic and abiotic factors (Gibb et al. 2015). At the community level, weighted means of functional traits represent the optimal survival strategies of species assemblages in the specific habitats (Muscarella and Uriarte 2016). In low-elevation habitats, interspecific competition can have marked effects in shaping the community (Machac et al. 2011). In high-elevation habitats, environmental filtering often selects species with specific functional traits that are suitable to the challenging environmental condition (Smith et al. 2014; Liu et al. 2018).

Therefore, functional diversity indices provide information on species function within communities and ecosystem function, while phylogenetic diversity indices capture evolutionary characteristics of community structure (Petchey and Gaston 2006; Swenson 2013). Species closely related in phylogeny are likely to be similar in functional aspects, but the patterns of phylogenetic and functional are not always consistent (Graham et al. 2009; Sobral and Cianciaruso 2016). Thus, functional and phylogenetic diversity provide complementary information that contributes to our understanding of insect community structure. If functional or phylogenetic diversity of a community is clustered, community assembly is primarily driven by environmental filtering; if functional or phylogenetic diversity is overdispersed, it is mainly driven by competitive exclusion; otherwise, it may be better explained by stochastic processes (HilleRisLambers et al. 2012). However, community assembly changes along elevational gradients are still incompletely understood for some insect taxa, particularly social insects like ants.

Ants are an ideal taxon for biodiversity research, as they are taxonomically diverse and widely distributed in terrestrial ecosystems (Hölldobler and Wilson 1990; Agosti et al. 2015). They play crucial roles in various ecological processes, acting as competitors, predators, scavengers, pollinators and seed-dispersers (Parker and Kronauer 2021). The ant assemblages are shaped by interspecific competition (Cerdá et al. 2013), habitat types (Nooten et al. 2019) and landscape composition (Dianzinga et al. 2020), and vary with environmental gradients such as aridity (Arnan et al. 2018) and elevation (Fontanilla et al. 2019; Silvestre et al. 2021), indicating that ants are highly sensitive to environmental changes. In old-growth forests, ants with smaller eyes, shorter scapes, narrower heads and shorter legs are able to nest in the leaf litter and humus layer (Nooten et al. 2021). To tolerate colder temperature and high levels of UV-B in high-elevation habitats, ants tend to have dark-colored and large bodies (Bishop et al. 2016). Up to now, the variation of ant functional

traits along elevations remains uncertain. Functional diversity shows a negative correlation with elevation (Fontanilla et al. 2019; Silvestre et al. 2021), while phylogenetic diversity presents either negatively correlated or uncorrelated (Machac et al. 2011; Reymond et al. 2013; Liu et al. 2018; Silvestre et al. 2021). Therefore, additional researches are required to gain a better understanding of ant diversity patterns in mountainous regions.

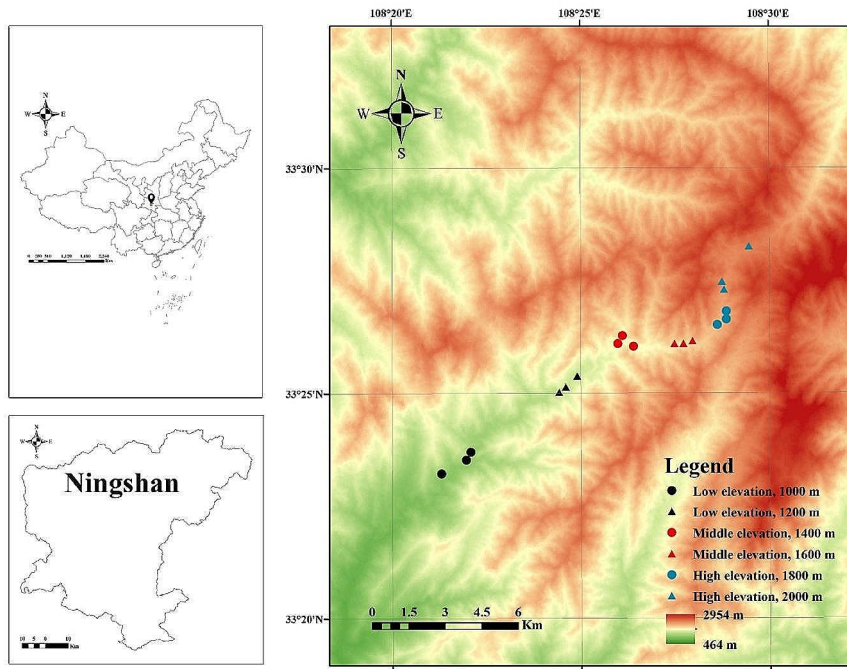
As a natural boundary between the warm temperate and subtropical zones, the Qinling Mountains, with vast elevational range and remarkably rich biodiversity, provide a unique opportunity to investigate how elevational gradients influence the ant diversity patterns (Hou et al. 2008; Yu et al. 2021). So far, studies conducted in this region have not yet addressed functional and phylogenetic diversity on ants (Wei et al. 2001; He et al. 2003). We performed field research of ant communities in six elevation bands in Ningshan County located in the Qinling Mountains. We tried to determine how the taxonomic, functional and phylogenetic diversity respond to elevations gradients, and analyzed how the functional trait distribution changed at different elevations. We hypothesized that (1) species and functional traits composition of ant communities among different elevations vary, and (2) indices of three aspects diversity decrease with elevations.

## Materials and methods

### Study area and sampling design

This study was conducted in Ningshan County (108°2'33" – 108°56'48"E, 33°7'11" – 33°50'38" N) located in Shaanxi Province, China, within the central area of the Qinling Mountains (Fig. 1). The Qinling Mountains are a major geographical dividing line in China, separating the northern and southern regions as well as the Palearctic and the Oriental realms (Wei et al. 2001). It is characterized by a northern subtropical semi-humid climate with an average annual temperature of 8–10 °C, an annual rainfall of 900–1200 mm and an average annual evaporation of 800–950 mm in the study area. The soil type mainly concludes mountain brown soil, dark brown soils and mountain meadow soil. The forest type is secondary, and the main dominant species are *Pinus armandii*, *Pinus tabulaeformis*, *Quercus aliena* var. *acuteserrata*, *Betula albosinensis*, *Tsuga chinensis*, *Picea wilsonii*, *Picea asperata* and *Abies fargesii* (Hou et al. 2008). The Natural Forest Protection Program (NFPP) has been applied in this region, which prohibits commercial logging of natural forests, implements forest management and mountain closure for the conservation of ecosystem and biodiversity. As parts of the dominant forest species in the region, *Pinus* species are widely distributed, which are pine forests or pine and oak mixed forests, and have an obvious vertical distribution pattern (Wei et al. 2014).

We established six study sites at different elevations, ranging from 967 to 2029 m.a.s.l. The elevational difference between adjacent sites was at least 200 m, and these sites consisted of 1000 (ES1), 1200 (ES2), 1400 (ES3), 1600 (ES4), 1800 (ES5) and 2000 m.a.s.l (ES6) (Table S1). Three plots were designed as replicates in each study site, for a total of 18 sampling plots, referred to as EP1–EP18 (Table S1). The sampling plots were spaced at least 300 m apart.



**Fig. 1** Location of the study area and sampling plots in Ningshan County, Shaanxi Province of China

## Ant sampling

We conducted this study in July 2021. At each sampling plot, we selected ten pine trees to establish a transect for collecting ants, with three to six meters between two adjacent pine trees next to each other (Fig. S1). To collect epigaeic ants, we set up three pitfall traps (diameter: 65 mm; depth: 77 mm; volume: 160 ml) filled with conserving liquid (ethanol and ethylene glycol, 1:1, approximately 50 ml) and one paper trap (baited with tuna and honey) using a 297 mm × 420 mm white paper. Three pitfall traps were arranged in a triangular shape around the base of each sampled pine tree, and one paper trap was set up near the base of each sampling pine tree. Collection by pitfall traps was performed for one week in total. During exposure, we inspected pitfall traps at intervals of three to four days, and poured the contents in the pitfall traps into a 200 mL plastic bottle. Then, the mixed liquid, consisting of ethanol and ethylene glycol as mentioned above, was added to every pitfall trap. Paper traps were left in the field for about 50 min, and then paper traps with ant samples were transferred into self-sealing bags.

To collect arboreal ants, we encircled every pine tree trunk with 10 cm wide gauzes baited with tuna and honey at a height of 1.3–1.5 m above the ground (Fig. S1). Trunk traps were left on the tree trunk for about 50 min, and then trunk traps with ant samples were placed into self-sealing bags. After that, two people collected ants by hand at 1.3–1.5 m of the tree trunk for 10 min. Ants collected by hand were transferred into 50 mL centrifuge

tubes. We brought all the ant specimens back to the laboratory, and preserved them in 95% alcohol for sorting and identification.

To identify specimens, we used a series of taxonomic keys and images available at [Ant-Web \(www.antweb.org\)](http://www.antweb.org) and [AntWiki \(www.antwiki.org\)](http://www.antwiki.org) to identify them to species or morphospecies. For specimens that were difficult to identify, we sent them to Guangxi Normal University for identification by ant taxonomists. We preserved all mounted and alcohol-preserved ant specimens in the College of Forestry, Northwest A&F University, China. Details on the species we collected were provided in Table S2.

## Functional traits

To quantify ant species functional traits, we selected six morphological characteristics: Weber's length (representing ant's body size), head width, scape length, eye length, mandible length and hind leg length (Table S3). Previous researches have shown that there exist relationships between the traits of a species and its diet, foraging pattern, and habitat preference (Yates et al. 2014; Gibb et al. 2015; Liu et al. 2016). To measure ant traits, we used an SDPTOP/SZN71 microscope with a digital camera connected to a computer that had an accuracy of at least 0.001 mm. Up to six workers of each species were randomly selected for trait measurements. In total, we measured 187 individuals representing 42 ant species, and we provided the details of these measurements in Table S4.

## Data analyses

At each sampling plot, the ant community data of arboreal and epigaeic ants collected using four methods were pooled together for statistical analyses. Species richness and sample coverage of ant communities at different elevations were calculated using 50 bootstrap replications with 95% confidence intervals based on Hill numbers with  $q=0$  in the package "iNEXT" (Chao et al. 2014). To illustrate the number of shared and non-shared species among different elevations, we produced a Venn diagram using the package "venn" (Dusa 2022). The abundance of ant community genera and subfamilies at different elevations was visualized using the R package "circlize" (Gu et al. 2014).

We performed clustering analysis based on species composition with Jaccard dissimilarity index using an unweighted pair-group method with arithmetic averages (UPGMA) by the R package "vegan" (Oksanen et al. 2022). Additionally, we clustered sampling plots based on the Euclidean matrix of community-weighted mean (CWM) values of functional traits using UPGMA. Furthermore, to analyze the distribution of ant communities and functional traits along elevational gradients, we visualized sample similarity using non-metric multidimensional scaling (NMDS) ordination based on Bray–Curtis dissimilarity distance using ant species composition and CWM values for each plot. We conducted a Permutation Multivariate Analysis of Variance (PERMANOVA) based on 999 permutations to determine statistical differences of ant communities among different elevations using the *adonis* function. We implemented this part using the "vegan" package.

Following the methods outlined in the literature (Liu et al. 2016; Nooten et al. 2021), we constructed our ant community phylogenetic tree based on the phylogenetic tree with 269 ant genera worldwide from Economo et al. (2018). Through comparison, ant species collected in this study that were not included in the phylogenetic tree were inserted using

the *add.species.to.genus* function in the “phytools” package (Revell 2012), and ant species not collected in this study were removed using the *drop.tip* function in the “ape” package (Paradis and Schliep 2019). Finally, we obtained an ant community phylogenetic tree containing 42 species, 27 genera, and four subfamilies for phylogenetic structure analysis. Phylogenetic diversity was calculated using Faith’s PD (PD, the sum of the branch length of species within a community; Faith 1992) and mean pairwise distance (MPD, the mean distance between pairs of species in the phylogenetic tree within a community; Webb et al. 2002). To generate an ant functional trait dendrogram, functional trait values were z-standardized and subsequently transformed into a Euclidean distance matrix. We clustered ant species based on the Euclidean distance matrix using UPGMA and calculated FD\_PD using the same method as PD (Petchey and Gaston 2006). Another functional diversity index was calculated as Rao’s quadratic entropy (Rao’s Q; Rao 1982).

To analyze the response pattern of ant diversity and functional traits among different elevations, we calculated ant species richness, Chao-1, Rao’s Q, FD\_PD, PD, MPD and CWM values. Chao-1, FD\_PD and the CWM values of Weber’s length, head width, scape length, mandible length and hind leg length passed the Shapiro–Wilk test for normality and the Bartlett test for variance heterogeneity. Thus, we analyzed them using one-way ANOVA with appropriate post hoc tests (Tukey HSD test) among different elevations. Furthermore, we employed the Kruskal–Wallis test (K-W test) with the Bonferroni method correction for multiple comparisons to compare species richness, Rao’s Q, PD, MPD and eye length CWM values among different elevations.

The standardized effect sizes (SES) of functional and phylogenetic diversity indices were measured by comparing observed values to expected values generated by null models ( $SES = (\text{Mean}_{\text{obs}} - \text{Mean}_{\text{null}}) / \text{sd}_{\text{null}}$ ) in the “picante” package (Kembel et al. 2010). Null model communities were generated by randomizing the community matrix through “independent swap” for 1000 times. Positive value represents overdispersion, and negative value indicates clustering. A two-sided Wilcoxon signed-rank test was used to assess the statistical significance of the observed functional and phylogenetic diversity and the expected indices of the null model.

All statistical analyses were performed in R version 4.2.1 (R Core Team 2022). We calculated Rao’s Q and CWM values by the *dbFD* function in the “FD” package (Laliberté and Legendre 2010). Tukey HSD test and K-W test were conducted using the “agricolae” package (de Mendiburu 2021). The NMDS with PERMANOVA test was performed by the “vegan” package. We compiled the heatmaps using the “pheatmap” package (Kolde 2019) and other figures using the “ggplot2” package (Wickham et al. 2016).

## Results

The rarefaction-extrapolation accumulation curves almost reached their asymptotes for all elevations, suggesting that the sampling was sufficient to analyze ant communities in the study area (Fig. S2a and b). A total of 11,347 ant specimens were collected in the field, which belonged to 42 ant species, 27 genera and 4 subfamilies (Table S2). Most species belonged to Myrmicinae (17 species), followed by Formicine (15 species), Ponerinae (6 species) and Dolichoderinae (4 species).

## The ant assemblages of species, genera, and subfamilies across different elevations

Ant species richness decreased with increasing elevations (Fig. S2a). Of all 42 ant species, *Lasius alienus*, *Nylanderia sharpie*, *Aphaenogaster caeciliae*, *Myrmica* sp., *Pheidole fervida* and *Temnothorax shannxiensis*, occurred in all elevations, and 14 species were only found in the plots of 1000 m. There was one species (*Nylanderia* sp.) only found at 1200 m and one (*Camponotus herculeanus*) only at 1600 m (Fig. S2c and Table S2).

At genus level (Fig. 2a and Table S5), the dominant species were from *Crematogaster*, *Lasius*, *Formica*, *Nylanderia* and *Pheidole* at 1000 m, respectively accounting for 21.88%, 16.88%, 15.62%, 10.72% and 10.12%. *Lasius* (39.27%), *Nylanderia* (30.24%) and *Aphaenogaster* (18.57%) dominated at 1200 m. The *Nylanderia* species were the most frequently collected at 1400 m (56.55%), followed by 1600 m (47.19%) and the least at 2000 m (0.58%). The *Prenolepis* species were found at 1400 m (2.03%), 1600 m (19.80%) and 1800 m (0.31%). The *Myrmica* species were abundant at 1800 m (50%) and 2000 m (81.19%).

At subfamily level (Fig. 2b and Table S5), Dolichoderinae group was only found at 1000 m (7.90%) and 1200 m (2.13%), and Ponerinae group was found at the plots under 1600 m. While, Formicinae and Myrmicinae group were found at all elevations. The collection frequency of Formicinae (73.27%) and Myrmicinae (88.29%) was the highest at 1200 m and 2000 m, respectively.

## The distribution and diversity of ant communities among different elevations

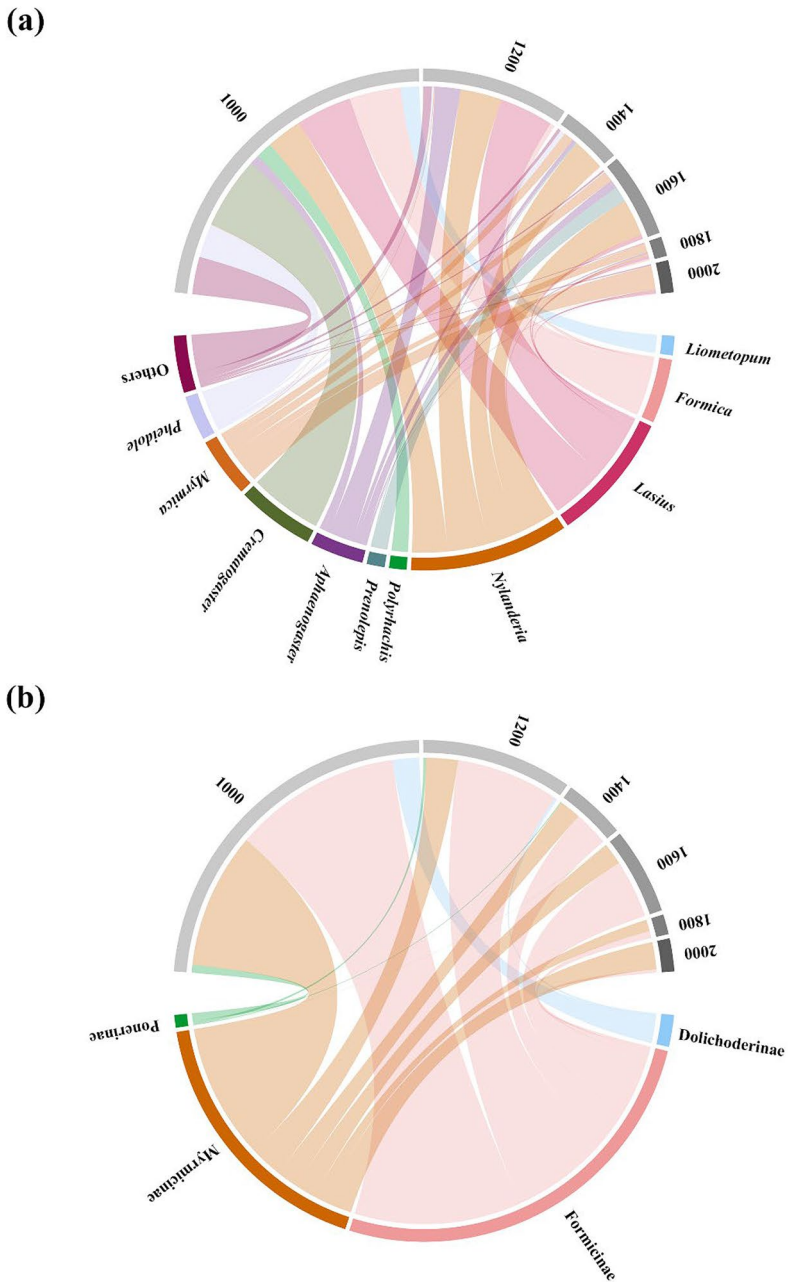
The Jaccard index-based cluster analysis (Fig. 3a) indicated that the ant communities of sampling plots at different elevations were clustered into three groups: 1000 m and 1200 m (low elevation), 1400 m and 1600 m (middle elevation), and 1800 m and 2000 m (high elevation). NMDS analysis showed that the ant community composition of 1400 m and 1600 m partially overlapped, revealing a distinct separation in ant communities between low elevation (1000 m and 1200 m) and high elevation (1800 m and 2000 m) (Fig. 3b,  $p=0.001$ ).

Significant differences were observed in all six diversity indices (Fig. 4 and Table S6). Multiple comparisons demonstrated that ant species richness, Chao-1, Rao's Q, FD\_PD, PD and MPD at low elevation were significantly higher than those at middle and high elevation, while there was no significant difference in all six indices between middle and high elevation.

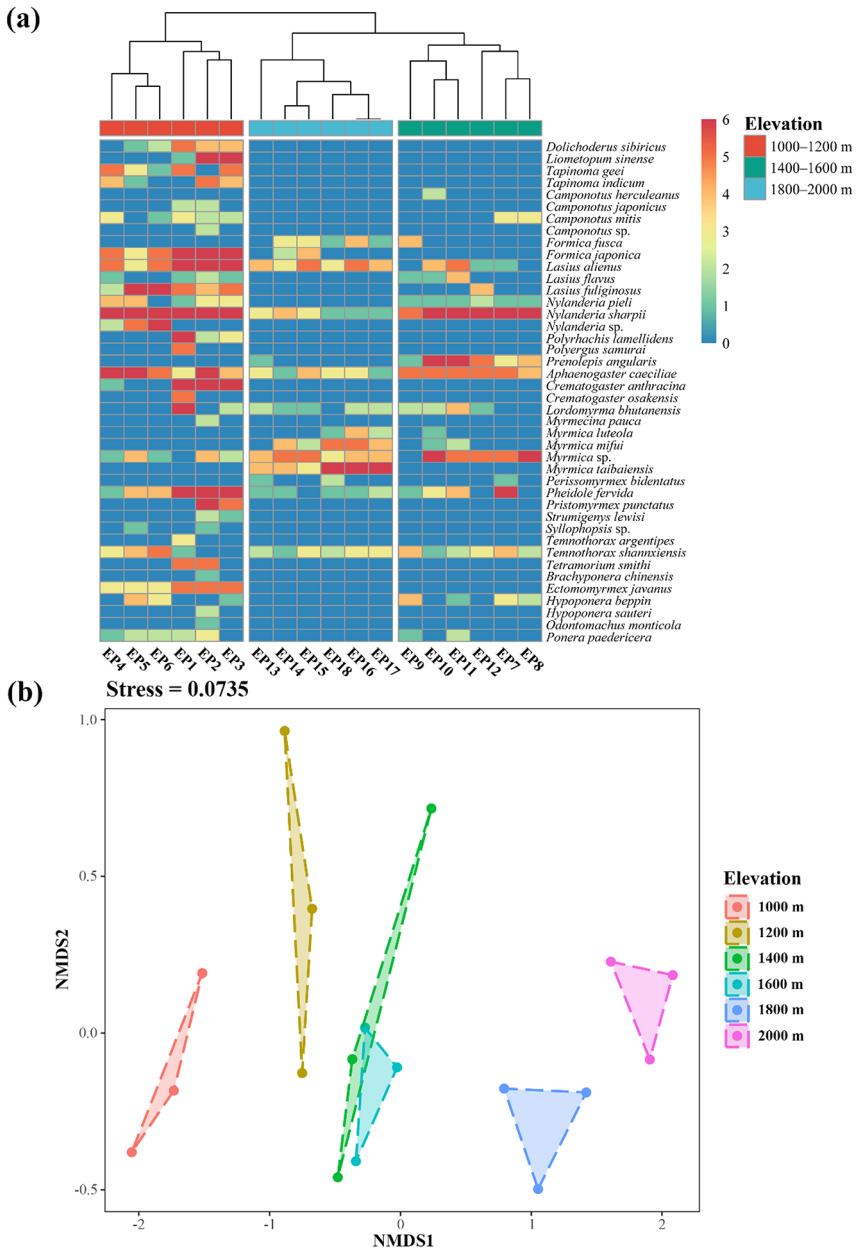
## Functional trait composition and divergence of ant communities among different elevations

Cluster analysis showed that the CWM values of ant communities at low elevation and high elevation were grouped together (Fig. 5a). According to the NMDS of CWM values, relative to middle elevation, functional trait composition of ant communities at low and high elevation was more similar (Fig. 5b,  $p=0.001$ ).

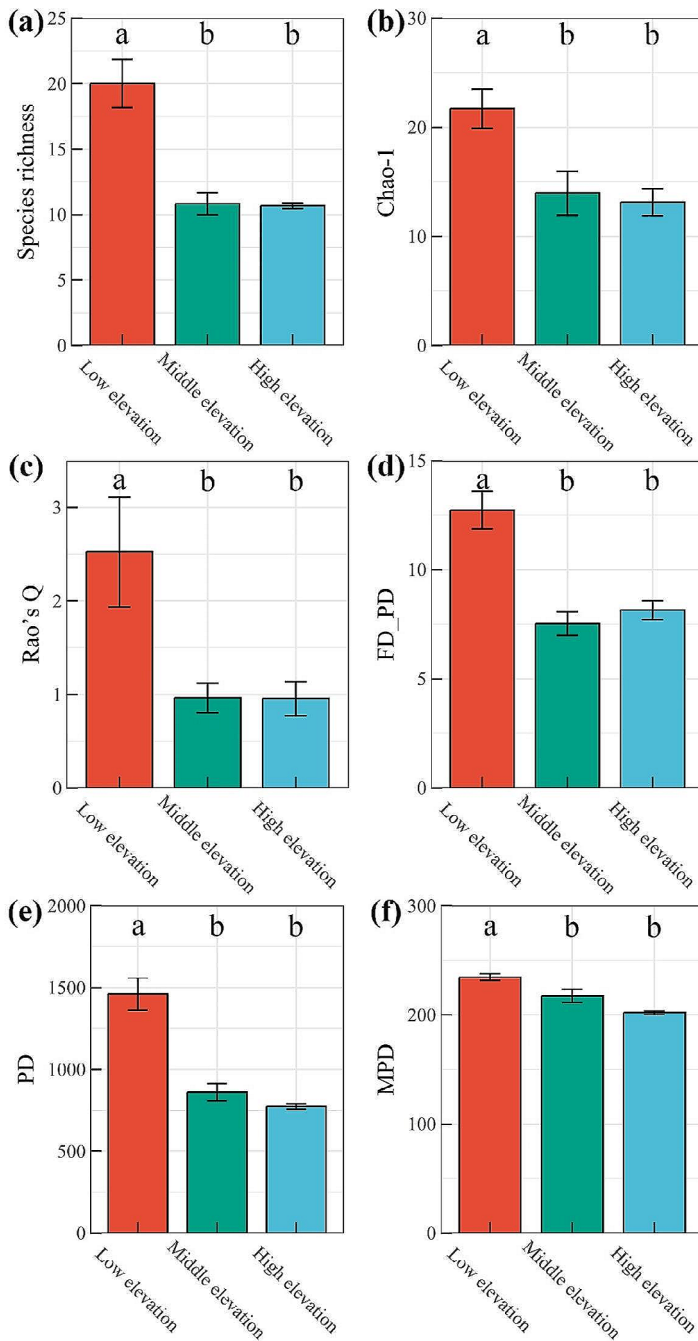
One-way ANOVA revealed significant variations in Weber's length, head width, scape length, mandible length and hind leg length among different elevations, and the K-W test indicated a significant difference in eye length among different elevations as well (Fig. 6 and Table S6). Regarding the CWM values of Weber's length, head width and mandible length,



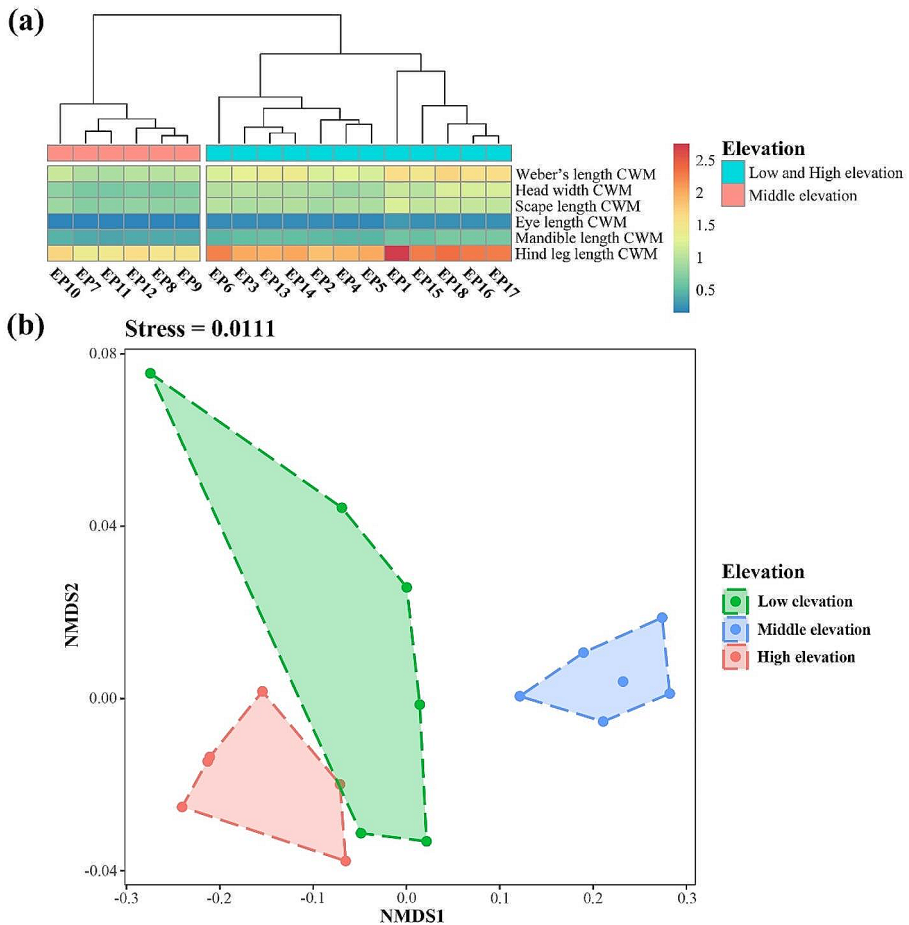
**Fig. 2** The abundance of genera (a) and subfamilies (b) of ant communities (lower half circle) at 6 elevations from 1000 m to 2000 m (upper half circle). The width of the links represents the abundance of genera and families at different elevations



**Fig. 3** (a) The heatmap of samples cluster based on Jaccard dissimilarity index of ant community composition using UPGMA; (b) Non-metric multidimensional scaling (NMDS) ordination based on the Bray-Curtis dissimilarity index of ant community composition. EP1–EP18 represents sampling plots, respectively. 1000–2000 m represents sampling sites at elevations of 1000–2000 m, respectively



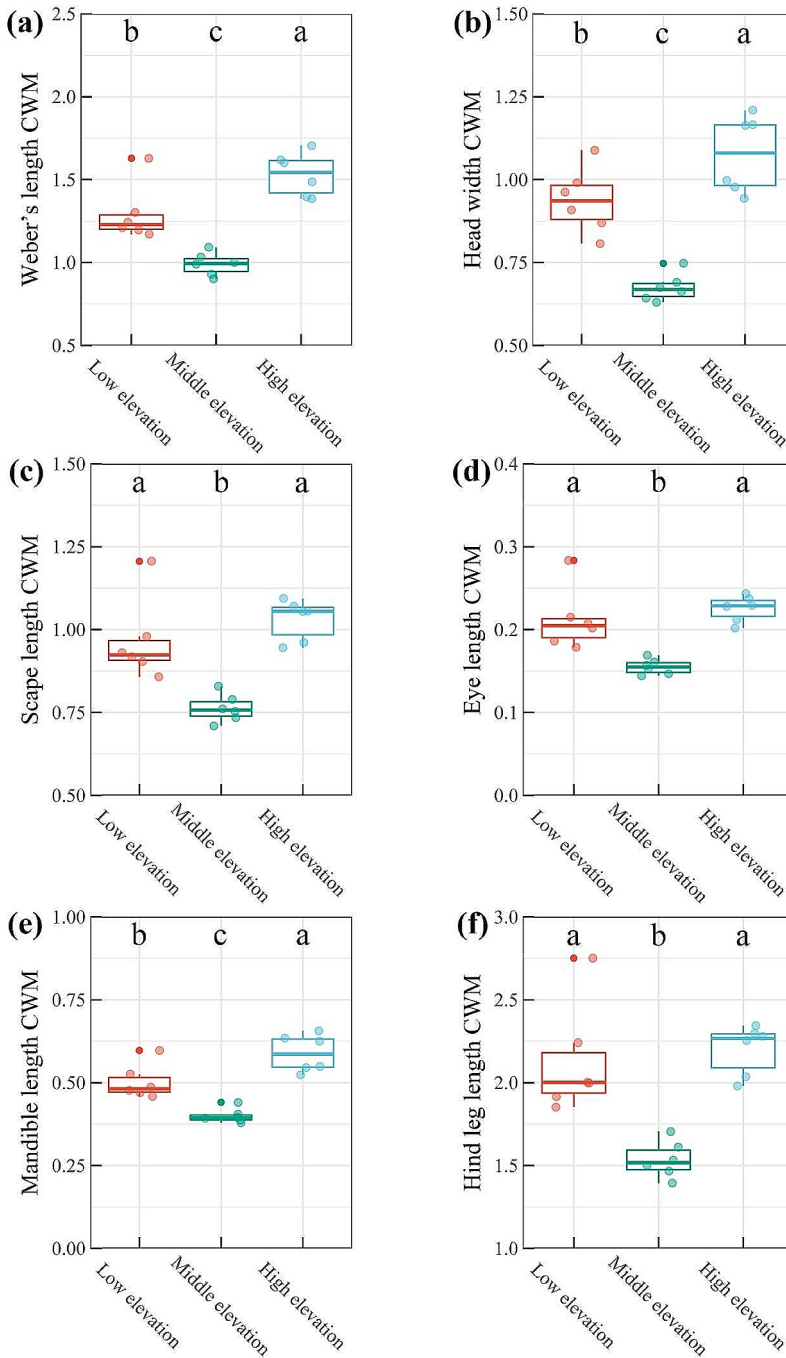
**Fig. 4** Bar charts showing taxonomic, functional and phylogenetic diversity at low, middle and high elevation. For taxonomic diversity, **(a)** Species richness and **(b)** Chao-1; For functional diversity, **(c)** Rao's quadratic entropy (Rao's Q) and **(d)** Faith's PD (FD\_PD); For phylogenetic diversity, **(e)** Faith's PD (PD) and **(f)** Mean pairwise distance (MPD). Showing mean  $\pm$  SE, and small letters: a and b indicating statistical significance. Significant level:  $p < 0.05$



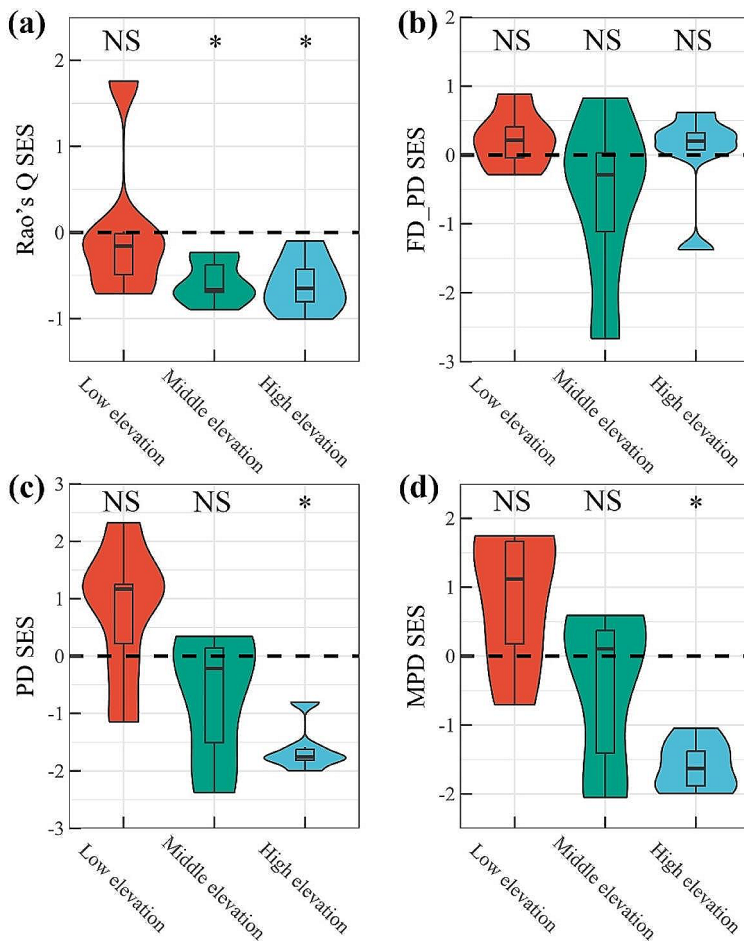
**Fig. 5** (a) The heatmap of samples cluster based on the euclidean matrix of community weighted mean (CWM) values using UPGMA; (b) Non-metric multidimensional scaling (NMDS) ordination based on the Bray-Curtis dissimilarity index of CWM of functional traits. EP1–EP18 represents sampling plots, respectively. Low elevation represents sampling sites at elevations of 1000 m and 1200 m. Middle elevation represents sampling sites at elevations of 1400 m and 1600 m. High elevation represents sampling sites at elevations of 1800 m and 2000 m

ant communities at high elevation were significantly larger than those at low and middle elevation, while ant communities at low elevation was also significantly larger than those at middle elevation (Fig. 6a, b and e). There was no significant difference in CWM values for scape length, eye length, and hind leg length between low and high elevation, but both were significantly higher than those at middle elevation (Fig. 6c, d and f).

For functional diversity, ant communities at middle and high elevation significantly deviated from null expectations using Rao's Q, with the SES less than 0 (Fig. 7a and Table S7). However, none of the ant communities diverged significantly from randomized ones in terms of FD\_PD (Fig. 7b). By contrast, both phylogenetic diversity indices were signifi-



**Fig. 6** Boxplots showing community weighted mean (CWM) values of functional traits at low, middle and high elevation. **(a)** Weber's length CWM; **(b)** Head width CWM; **(c)** Scape length CWM; **(d)** Eye length CWM; **(e)** Mandible length CWM; **(f)** Hind leg length CWM. The box represents 25–75th percentiles and whiskers extending to 1.5 times the interquartile range. Small letters a, b and c indicate statistical significance. Significant level:  $p < 0.05$



**Fig. 7** Violin plots showing kernel probability density of the standardized effect sizes (SES) of functional and phylogenetic diversity at low, middle and high elevation. For SES of functional diversity, (a) SES of Rao's quadratic entropy (Rao's Q SES) and (b) SES of Faith's PD (FD\_PD SES); For SES of phylogenetic diversity, (c) SES of Faith's PD (PD SES) and (d) SES of mean pairwise distance (MPD SES). The box inside represents 25–75th percentiles and whiskers extending to 1.5 times the interquartile range. Asterisks (\*) represents significant and NS representing not significant between the observed and null distribution of phylogenetic and functional diversity. Significant level:  $p < 0.05$

cantly lower than random expectations at high elevation, but not at low or middle elevation (Fig. 7c and d).

## Discussion

Overall, we found that ant diversity of taxonomic, functional and phylogenetic aspects showed a decreasing trend with increasing elevations in the Qinling Mountains. Ant assemblages varied along elevations according to the taxonomic group to which they belong.

Functional traits of ants were more similar between those found at low and high elevation, whereas ants living at the middle elevation have shorter Weber length, narrower heads, shorter scapes, smaller eyes, shorter mandibles and shorter hind legs. Remarkably, both functional and phylogenetic diversity was clustered at high elevation. The community functional diversity at middle elevation was clustered, while phylogenetic diversity did not diverge. Neither functional nor phylogenetic diversity at low elevation diverged.

Supporting hypothesis one, our findings suggested that the ant community species composition at low elevation (1000 m and 1200 m), middle elevation (1400 m and 1600 m), and high elevation (1800 m and 2000 m) differed clearly from each other. It is well-established that mountain ecosystems undergo a significant temperature decrease as elevation increases (Rahbek et al. 2019). Being highly thermophilic insects, the variation in ant species composition might be related to the capacity of species to match their thermal tolerance range to temperature profile in the mountain (Bishop et al. 2017; Joseph et al. 2019). Six species were collected at all elevations, suggesting they were generalist species with a broad temperature adaptation and a wide range of distribution. *Myrmica* species dominated the community at high elevational regions, and this is consistent with previous studies (Wei et al. 2001; Liu et al. 2018). Leahy et al. (2022) shows that highland ants have a smaller tolerance to high temperatures compared to lowland ants. Therefore, we speculated that the *Myrmica* genus population is susceptible to negative impacts from global climate warming.

In high elevational habitats, there are extreme ecological adaptations because of poor resource availability and restrictive environments (Thakur and Chawla 2019). Plant productivity has effects on ant communities (Silvestre et al. 2021), and it may be related to the honeydew secretions of hemipterans and the sugary liquids from plants' extrafloral nectaries (Parker and Kronauer 2021). Many Ponerine ants are specialist predators on soil arthropods (Wilson and Hölldobler 2005), and these ants were not found at high elevation. With increasing elevations, the soil becomes less nutrient-rich and the prey are not abundant (Xie et al. 2022), potentially restricting the occurrence of Ponerinae ants, which are mostly predator ants. Our research showed that elevation had a certain impact on the formation of ant communities.

In this study, the diversity indices of three aspects showed a decreasing trend with the increase of elevations (supporting hypothesis two). This finding regarding taxonomic diversity is consistent with many previous studies (de Castro et al. 2020; Fontanilla et al. 2019; Silvestre et al. 2021; Wei et al. 2001), and this phenomenon might be associated with decreases in temperatures and changes in vegetation types with increasing elevations in the Qinling Mountains (Wei et al. 2001). Warmer habitats with high productivity tend to harbor more species than cooler ones (Silvestre et al. 2021). Combining the trends of functional and phylogenetic diversity we observed, it implied that the extreme conditions at high elevation are challenging for ants.

Our study also verified the first hypothesis, showing that functional traits of ant communities varied at different elevation. As Bishop et al. (2016) found, ant species at high elevation have longer Weber's length (larger body). Additionally, ant functional traits showed wider heads, bigger eyes, longer scapes, longer mandibles and longer hind legs. Longer Weber's length improves the ability for species to thermoregulate and reduce heat loss in cold environments (Bishop et al. 2016). Longer scapes and bigger eyes enhance the capability of foraging for food, especially in resource-poor habitats (Parr et al. 2017). Longer hind legs are positively related to their mobility and efficiency (Weiser and Kaspari 2006). Wider

heads and longer mandibles might broaden their food spectrum, and these physical advantages would help them survive in environments where food is scarce (Gibb et al. 2015). It is noteworthy that ant communities at high elevation were clustered in both functional and phylogenetic dimensions, suggesting that the ant community assembly in these habitats was influenced by environmental filtering. This also indicated that different ant species from the closely phylogenetic clades, possessing similar functional traits that contribute to adaptation to extreme environments, persist in high-elevation habitats (Machac et al. 2011; Smith et al. 2014; Liu et al. 2018).

Interestingly, there were shorter Weber's length, narrower heads, shorter scapes, smaller eyes, shorter mandibles, and shorter hind legs of ant communities at middle elevation. We speculated that these might be connected with the forest microhabitat, and the ant assemblages here were primarily composed of species which forage in leaf litter layer. The size-grain hypothesis suggests that as habitats become less planar and more rugose, the body size of ants tends to decrease (Kaspari and Weiser 1999). Due to the distribution of pine forests or pine and oak mixed forests at middle elevation (Wei et al. 2014), a large amount of pine needles accumulates on the ground. These functional traits were convenient for ants to pass through pine needles to forage and nest under the leaf litter. At middle elevation, only functional diversity clustering was observed, implying there was environmental filtering on ant functional traits.

Low-elevation areas, characterized by abundant food resources and diverse habitats, provide numerous niches and facilitate niche specialization (Nooten et al. 2019; Dianzinga et al. 2020). Our results demonstrated the predominant role of stochastic assembly processes in shaping ant communities at low elevation, implying that species with equivalent competitive abilities might have comparable opportunities to occupy available niches (HilleRisLambers et al. 2012). In contrast, Machac et al. (2011) and Liu et al. (2018) considered interspecific competition as the primary factor in structuring ant communities. Interspecific competition is indeed one of the major factors in shaping ant communities (Cerdá et al. 2013). In lowlands, due to limited territories in the canopy, there is intense competition among arboreal ant species which have similar nesting habits, activity periods, and recruitment rates (Camarota et al. 2016). In addition to the arboreal ants, ants in the litter also contribute to low-elevation ant community composition, and the competition among ant colonies is less in the litter than in the canopy (Yanoviak and Kaspari 2000). We collected both arboreal and epigeic ant species in our study, and this might diminish the competition we observed. Furthermore, although conditions in lowland habitats are not harsh, the attributes of these habitats still affect ant communities. Ribas et al. (2003) found that high diversity and availability of resources in a habitat increase ant species richness for both specialist and generalist species. The mechanism of ant community assembly in low-elevation habitats might be complex and worthy of our further study in the future. The values of six ant functional traits at low elevation were all significantly greater than those at mid elevation, and the functional trait composition of ant communities at low elevation was similar to that at high elevation. This could be attributed to the various niches in low-elevation habitats, where ants with different foraging strategies and dietary preferences are distributed simultaneously and widely. For instance, at low elevation, surface foragers have longer Weber's length, broader heads, longer scapes, bigger eyes and longer hind legs, whereas litter foragers display contrasting trait characteristics (Weiser and Kaspari 2006; Gibb et al. 2015).

In conclusion, we have revealed how ant diversity pattern responded to elevation in the Qinling Mountains of China. Both stochastic processes and environmental filtering were involved in ant community assembly in the study regions. Given the variation in functional traits of ant communities along the elevational gradients, ant species may have an optimal elevational range suitable for themselves. Therefore, there may be little possibility for some ant species to disperse to habitats at other elevations where they are not currently distributed. In the context of global warming, the distribution of some species is shifting to higher elevations, while most of these may be generalist species (Menéndez et al. 2006; Chen et al. 2009), and specialist ant species, especially those living in high mountain regions, will be the focus of our conservation efforts in the future.

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## Declarations

**Competing interests** The authors have no competing interests to declare that are relevant to the content of this article.

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